



Soknarith (Soki) Sem

☎ (401) 952-6819 — ✉ seminaia401@gmail.com — 🌐 [Seminaia](#) — in [soknarith-sem](#) — 📍 Worcester, MA

SUMMARY

Materials engineer with experience in process optimization, manufacturing support, material characterization, and computational analysis across research and industrial environments. Background includes cGMP operations, statistical process control, root cause investigation, SOP development, and atomistic simulation. Brings a combined process and materials perspective to improving manufacturing efficiency, process reliability, and product quality.

WORK EXPERIENCE

Research Assistant & Graduate Teaching Assistant *Worcester Polytechnic Institute - Worcester, MA*

May 2024 to Present

- Quantified defect energetics and transport behavior in oxide perovskites using DFT and thermodynamic modeling to evaluate material stability and performance.
- Analyzed structure-property-process relationships across temperature and chemical potential regimes to understand defect behavior relevant to semiconductor manufacturing.
- Computed temperature-dependent free energies and defect formation energies using Monte Carlo methods to predict equilibrium defect concentrations and transport properties across operating conditions.
- Assisted in teaching Metal Additive Manufacturing, Introduction to Materials Science and Engineering, Advanced Thermodynamics, and Applied Analytical Methods by grading assignments, supporting students in technical problem solving, and reinforcing core engineering concepts.
- Led help sessions and office hours to support student understanding of course material. Helped with homework assignments and exam preparation by providing guidance on problem-solving approaches and clarifying technical concepts.

Chemical Process Engineer

May 2023 to May 2024

Knowles Precision Devices - New Bedford, MA

- Led R&D-to-production transition for high-voltage capacitor dielectric fluids, supporting scale-up and manufacturing readiness.
- Optimized manufacturing layout and process flow using LEAN principles and statistical process controls, reducing production costs by 15%.
- Authored SOPs and work instructions for material handling and process scale-up, for training technicians and operators.
- Performed root cause analysis on material-related issues, including deviation investigation and CAPA implementation.

R & D Technician

Mar 2023 to May 2023

Factorial Energy - Woburn, MA

- Fabricated Li-metal batteries in controlled glovebox and dry-room environments to support researchers in evaluating battery performance under different conditions.
- Reduced material waste by 20% through inventory system redesign.

Downstream Purification Engineer

Jul 2022 to Mar 2023

Eurofins Scientific - Framingham, MA

- Downstream purification of therapeutic proteins, with experience in chromatography and filtration techniques at the lab, pilot, and manufacturing scale.
- Purified therapeutic proteins using AKTA Avant systems with Unicorn software achieving 95-99% purity.
- Purification of antibodies and enzymes using various chromatography and filtration techniques.
- Applied DOE methods to optimize buffer conditions and flow rate for maximum yield and purity at the lab-scale.
- Performed lab-scale membrane filtration and chromatography experiments to support pilot scale, then scaled up to manufacturing scale.
- Characterized purified proteins using SDS-PAGE, HPLC/MS, GC/MS, UV-Vis, and other analytical techniques to confirm purity and identity.

EDUCATION

M.S. Materials Science & Engineering

Worcester Polytechnic Institute - Worcester, MA

GPA: 3.50
2024 to 2026

Relevant coursework: Advanced Thermodynamics, Advanced Process Engineering, X-Ray Diffraction

B.S. Chemical Engineering

University of Rhode Island - Kingston, RI

GPA: 3.16
2019 to 2022

Minors: Chemistry

B.S. Applied Mathematics

University of Rhode Island - Kingston, RI

GPA: 3.16
2019 to 2022

Minors: Physics

TECHNICAL SKILLS

Atomistic Simulation: DFT (VASP & GPAW), Molecular Dynamics (LAMMPS)

Scientific Computing: Python, MATLAB, ASPEN Plus Dynamics, Linux, LaTeX, R, Unicorn

Laboratory: Glovebox, clean-room, HPLC, AKTA FPLC, TFF, SEM, XRD, DSC/TGA, UV-Vis, Osmometry, HPLC/MS, GC/MS, pH-meter, SDS-PAGE, KF titration meter

Manufacturing: SPC, DOE, Lean/Six Sigma, Process optimization, SOP development, Tech Transfer (MSAT), cGMP